

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (CL) Examination

May 2014

CL 214 Fluid Mechanics-II

Date: 07.05.2014, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Explain boundary layer separation and its control in details. Also give various applications where separation occurs. [07]
- Q - 2 (a) What are the criteria for most economical trapezoidal section and the best slope that could be designed for it [03]
- (b) Derive the expression for different flow corrections in gradually varied flow in an open channel. [05]
- (c) In a rectangular channel of 0.5 m width, a hydraulic jump occurs at a point where depth of water flow is 0.15 m and Froude number is 2.5. Determine : (1) The specific energy (2) The critical and subsequent depths (3) Energy dissipated. [06]

OR

- Q - 2 (a) Explain specific energy and specific energy curve. [05]
- (b) Determine the most economical section of a rectangular channel carrying water at the rate of $0.5 \text{ m}^3/\text{sec}$; the bed slope of the channels being 1 in 2000. Take chezy's constant $C = 50$. [05]
- (c) Derive the expression for the depth of hydraulic jump. [04]
- Q - 3 (a) Show that the maximum velocity of flow through a circular pipe is 2 times the average velocity. [05]
- (b) What is kinetic Energy correction factor and momentum correction factor. Derive expression for either of the two. [04]
- (c) A pipe 240 mm in diameter and 10000 m long is laid at a slope of 1 in 180. An oil of specific gravity 0.85 and viscosity 1.5 poise is pumped up at the rate of $0.02 \text{ m}^3/\text{sec}$. Find (1) Head lost due to friction (2) Power required to pump the oil. [05]

OR

- Q-3 (a) An oil of viscosity 9 poise and specific gravity 0.9 is flowing through a horizontal pipe of 60 mm diameter. If the pressure drop in 100 m length of the pipe is 1800 KN/m^2 , determine (1) the rate of oil (2) the centre line velocity (3) the total frictional drag over 100 m length (4) the power required to maintain the flow (5) the velocity gradient at the pipe wall (6) the velocity and shear stress at 8 mm from the wall. [08]
- (b) Derive expression for the couette flow. [06]

SECTION - II

- Q-4 (a) 1. Specific Energy of flowing water through a rectangular channel of width 5 m is ---- [06]
 ----, when the discharge is $10 \text{ m}^3/\text{s}$ and depth of water is 3 m.
 2. Maximum velocity through a circular channel of diameter 'D' is ----.
 3. Criteria for hydro-dynamically smooth and rough boundaries are ----, ----
 ---- and ----, where k is average ht of irregularities and δ' is thickness of laminar sub layer.

- (b) 1. Prandtle considers mixing length as ---- [01]

- Q-5 (a) Derive equation for velocity of sound wave $C = \sqrt{\frac{dp}{d\rho}}$ [07]

- (b) A supersonic aeroplane is flying at 1500 km/hr at an altitude of 10 km above sea level in standard atmosphere. The pressure and density are as 2.5 N/cm^2 (absolute) & 0.4 kg/m^3 . Calculate the pressure, temperature & density at the stagnation point on the nose of plane. $R=0.287 \text{ KJ/kgK}$ and $\gamma=1.4$ [07]

OR

- Q-5 (a) Explain propagation of pressure waves in compressible fluid for different Mach numbers with proper diagram. [07]

- (b) Calculate the velocity & Mach number of a supersonic aircraft flying at an altitude of 1200 mt where the temperature is 0°C . The sound of the air craft is heard 3 seconds after the passage of air craft over the head of an observer. $R=287 \text{ J/kgK}$ and $\gamma=1.4$ [07]

- Q-6 (a) Discuss different types of similarities between the model and prototype. [05]

- (b) Explain following terms:- 1. Froude number 2. Weber number [02]

- (c) The efficiency η of a fan depends on density ρ , dynamic viscosity μ of the fluid, angular velocity ω , diameter D of the rotor and the discharge Q. Express η in terms of dimensionless parameters. [07]

OR

- Q - 6 (a) Discuss different types of models used in model similarity laws. [04]
- (b) In 1 in 40 model of a spillway, the velocity and discharge are 2 m/sec and 2.5 m³/sec. [03]
Find the corresponding velocity and discharge in the prototype.
- (c) Derive on the basis of dimensional analysis suitable parameters to present the thrust [07]
developed by a propeller. Assume that the thrust T depends upon the angular velocity ω ,
speed of advance V , diameter D , dynamic viscosity μ , mass density ρ and elasticity of
the fluid medium which can be represented by the speed of sound C in the medium.